

My Answers to

Random Questions I Think About

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Write-ups of random questions I've thought about and tried to understand more deeply. Started doing this at the end of May 2026. In the age of LLMs, there is no excuse for not understanding anything that you work with as deeply as you'd like.

I'll note that most questions that I ask here are very basic, as that's how I tend to operate. Unless I understand something in its most natural setting, I am confused. Asking basic questions is largely how I get unconfused.

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1 Basic physics

1.1 Statistical mechanics

What is an ensemble?

(to-do)

Why do physical systems minimize their free energy?

We'll show this using the saddle-point argument. Consider a partition function,

$$Z(\beta) = \int dE \rho(E) e^{-\beta E} = \int dE e^{S(E) - \beta E}.$$

If the system is large ($N \sim 10^{23}$), it is well-approximated by its saddle-point value if S and E are extensive quantities,

$$Z(\beta) \approx e^{\max(S(E) - \beta E)}.$$

(need to completely justify this claim) Looking more closely at the quantity in parentheses gives

$$\max(S(E) - \beta E) \iff \min(\beta E - S) \iff \min(E - TS) =: \min F.$$

So, it's overwhelmingly likely¹ that a system will take a value that minimizes its free-energy, $F := E - TS$.

Notice that the two assumptions in this model beget some of its failure modes: we assumed that $N \sim 10^{23}$ and that E and S were extensive quantities in N . (exampels of where this doesn't hold)

□

What is the chemical potential?

(to-do)

¹The probability that a system takes a non-equilibrium free energy value is on the order of $e^{-10^{23}}$. This is comically tiny.

2 Basic math

2.1 Analysis

Big-O notation

(finish)

Useful fact about Fourier transforms: the uncertainty principle

Statement from Hannah Cairo: (see the talk [here](#))

Idea 2.1 (The uncertainty principle)

If a function f has support on an interval $[-R, R]$, then its Fourier transform $\tilde{f}$ is essentially constant on $[-1/R, 1/R]$.

This generalizes to boxes (is essentially a special word here? i'd prefer to use approximately)

Idea 2.2

If $\text{supp } f \subset [-A, A] \times [-B, B] \times [-C, C]$, then $\tilde{f}$ is essentially constant on boxes of $[-1/A, 1/A] \times [-1/B, 1/B] \times [-1/C, 1/C]$.

We can easily draw this idea to get some intuition for the result:

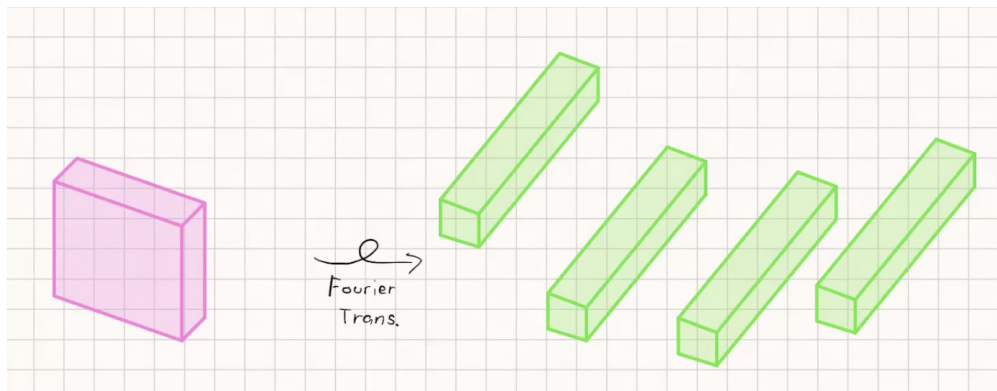


Figure 1: Support of f on the left, and (support ?) of Fourier transform on the right.

Notice the duality between small R and big $k \sim 1/R$, and vice-versa—this is reflected in the dimensions of the boxes on the left and right-hand side above. The reason why there are multiple boxes is because (not sure honestly. how does the support of a function fourier transform? this is a useful question to answer for physical considerations as well! i.e. $f(x)$ has some supp, wb $\tilde{f}(k)$?)

3 Physics topics

3.1 Condensed matter

What is a Fermi surface?

We'll start with an example. Take a fermionic gas in a cube with side length L . Its possible energies are

$$E_{\mathbf{k}} = \frac{\hbar^2 \mathbf{k}^2}{2m} = \frac{\hbar^2 \mathbf{n}^2}{2mL^2}.$$

Since these particles are fermions, they obey the Pauli exclusion principle. Thus, if we fill the states starting from the ground state upwards, we **(nope. don't understand why μ even exists or is a thing)**

3.2 Quantum field theory